

MLCB TEAM PROJECT

Communication-Aware Machine Learning for Major Depressive Disorder

Testing a microglia → neuron communication axis from single-nucleus RNA-seq of the human prefrontal cortex

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Re-analysis of Maitra et al. (2023), Nature Communications 14, 2912

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BACKGROUND

Two cell types, two sexes — reported separately

Major depressive disorder (MDD) has a poorly resolved molecular substrate in the human brain. Single-nucleus RNA-seq of the dorsolateral prefrontal cortex lets us ask *which cell types* carry the disease signal.

Maitra et al. (2023) profiled ~160,000 nuclei from 71 donors and found the strongest case/control differences in:

FEMALES

Microglia (cluster Mic1)

MALES

Deep-layer neuron (ExN10_L46)

Presented as two distinct, sex-specific results.

OUR HYPOTHESIS

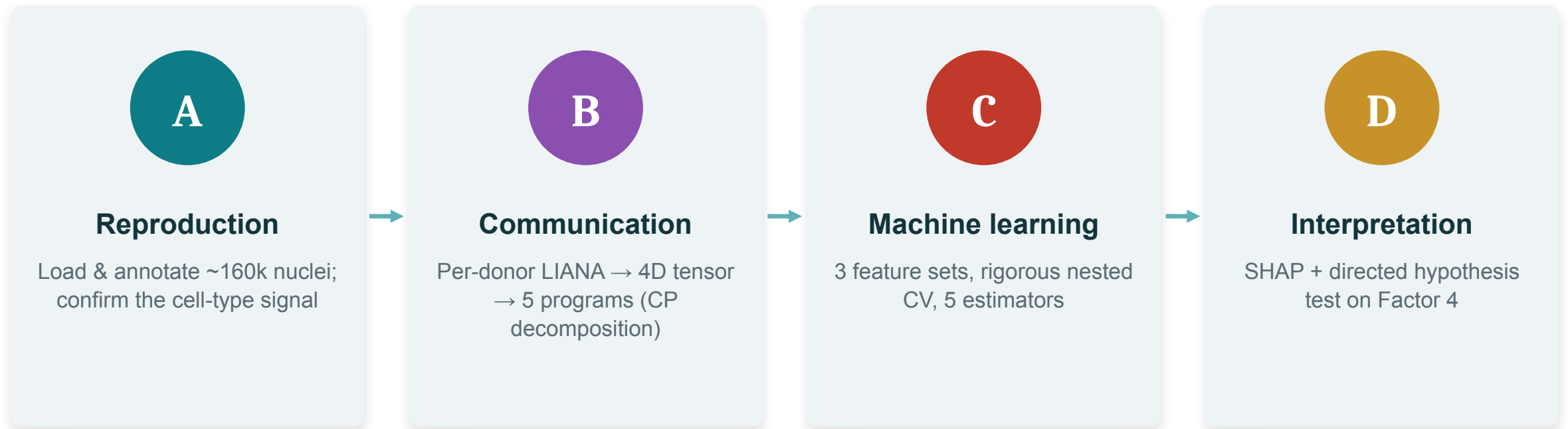
Not two findings — one axis



The apparent sex-specificity is partly an artefact of **per-cluster differential expression**, which treats each cell type in isolation and cannot see a signal that lives *between* cell types.

APPROACH

An end-to-end, checkpointed pipeline



Each expensive step is checkpointed — a runtime timeout never costs more than one step.

DATA

71 donors — and one confound that shaped everything

71

donors (the CV unit)

37 / 34

MDD / Control

38 / 33

female / male

160,711

high-quality nuclei

THE CONFOUND TO TRACK

Sex is perfectly confounded with cohort/library: females = GSE213982, males = GSE144136.

Any train-on-one-sex / test-on-the-other split is also a train/test **across-batch** split — so a failure to generalise could be batch, not biology.

This single fact reshaped the entire evaluation design (no leave-one-sex-out).

Confirm the cell-type signal, leakage-aware

Annotations already validated

We do not re-cluster (stochastic); the authors' QC and cell-type labels are reused, baked into each cell ID.

DEG sanity check passes

Mic1 (female): synaptic/adhesion genes up. ExN10_L46 (male): metabolic/synaptic genes (GAPDH, PKM, CKB, VAMP2, SNCB) down.

HONEST CAVEAT

Nucleus-level Wilcoxon *pseudoreplicates* — nuclei from one donor are not independent, inflating “significant” DEG counts.

Fix: all definitive donor-level statistics downstream use **pseudobulk** — exactly why the authors did too.

From ligand–receptor scores to 5 programs

1

Per-donor LIANA

`rank_aggregate.by_sample` gives each of the 71 donors its own L–R scores (not per-condition, which would be degenerate).

2

4D tensor

71 donors × 1650 L–R pairs × 8 senders × 8 receivers. Masked slices left as NaN (borrow strength, not impute).

3

–log₁₀ magnitude

`magnitude_rank` (small = strong) transformed so large = strong, matching LIANA's convention.

4

Non-negative CP

PARAFAC, elbow-selected rank 5 → five communication programs; a donor's loading = how active that program is.

Methodology shift: *LIANA (Python) instead of CellChat (R) — native per-donor inference, no fragile rpy2 setup.*

One program is diagnosis-leaning and cohort-clean

Factor	Program (sender → receiver)	p(diagnosis)	p(sex/cohort)	note
1	ExN → ExN	0.348	0.778	<i>no signal</i>
2	ExN → Ast/OPC	0.398	0.0025	<i>cohort/batch</i>
3	OPC/Ast → mixed	0.479	0.0002	<i>cohort/batch</i>
4	End + Mic → ExN10_L46	0.057	0.108	hypothesis program
5	mixed → End	0.886	0.00002	<i>cohort/batch</i>

Why Factor 4: the microglia-sender × ExN10_L46-receiver product is $0.40 \times 0.54 \approx 0.22$ — far above any other factor. Selected *a priori by biology*, not by p-value. The batch confound self-segregates into Factors 2/3/5.

Three feature sets, one decisive comparison

Expression-only

Per-cell-type pseudobulk (logCPM). The baseline.

Communication-only

The 5 factor loadings.

Combined

Pseudobulk + 5 factors. The decisive test.

Communication is computed from expression (a ligand's “signal” is its gene expression), so the two are not independent — which is exactly why the **combined comparison is the decisive test** of whether communication adds anything.

Validation: repeated nested CV (3×5×3, 20 Optuna trials), 5 estimators, bootstrap 95% CIs. **Leakage rules:** CV unit = donor (M24 merged); all feature selection fit in-fold; **tensor re-fit inside each fold** for the headline numbers.

Expression wins; combined never separates

Estimator	Communication	Expression	Combined
LR	0.633	0.646	0.694
GNB	0.633	0.643	0.673
LDA	0.633	0.667	0.653
RF	0.536	0.688	0.667
XGB	0.490	0.714	0.714

Median AUC, rigorous nested CV. Communication-only: linear models ≈ 0.633 , trees at chance.

0.714 → 0.714

best model (XGB): identical with or without factors

VALUE-ADD (combined – expression)

Positive on average but **not CI-separated**: LR +0.048, GNB +0.031; LDA/RF slightly negative; XGB flat.

Honest read: communication carries weak real signal (linear models only) but adds **no statistically demonstrable value** beyond expression at $n = 71$.

Higher in MDD donors of both sexes

0.632

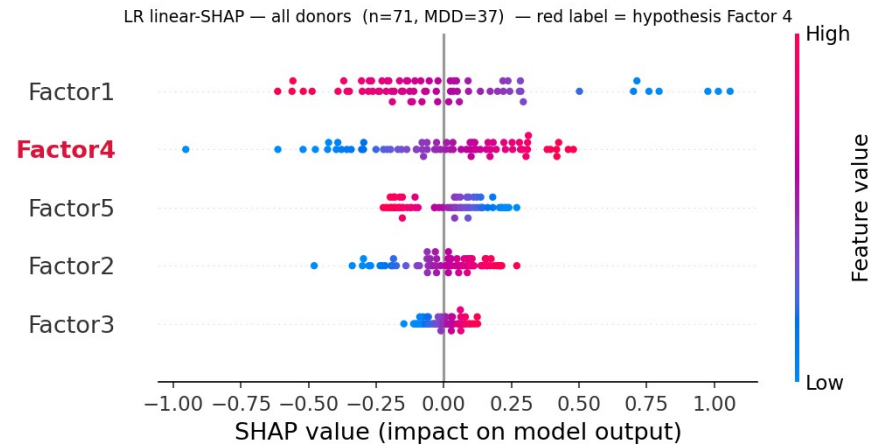
AUC · overall

0.702

AUC · males · p=0.05

0.578

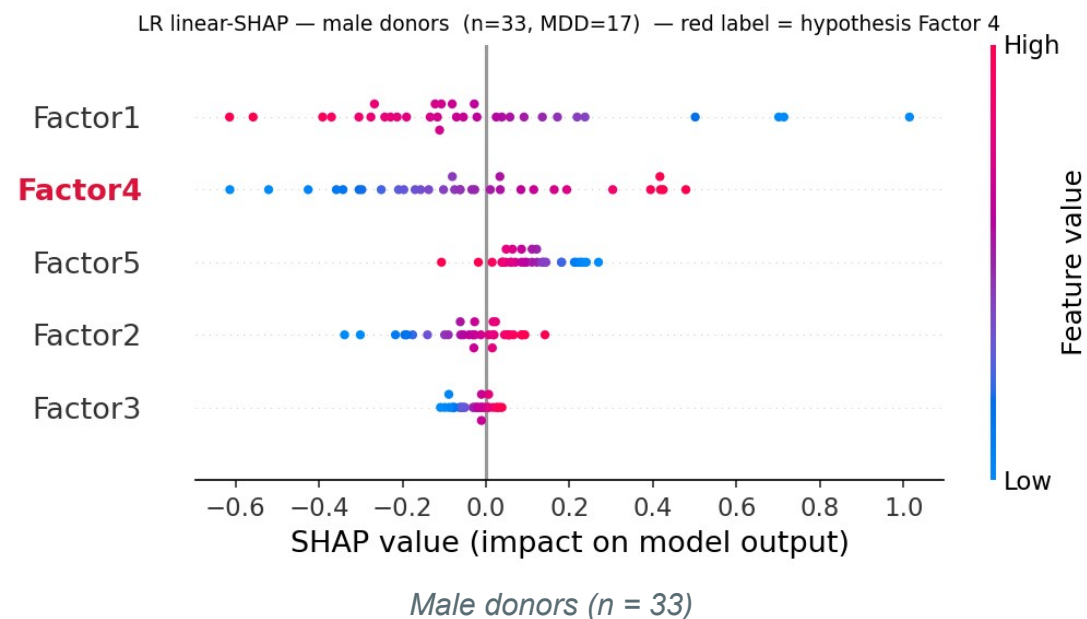
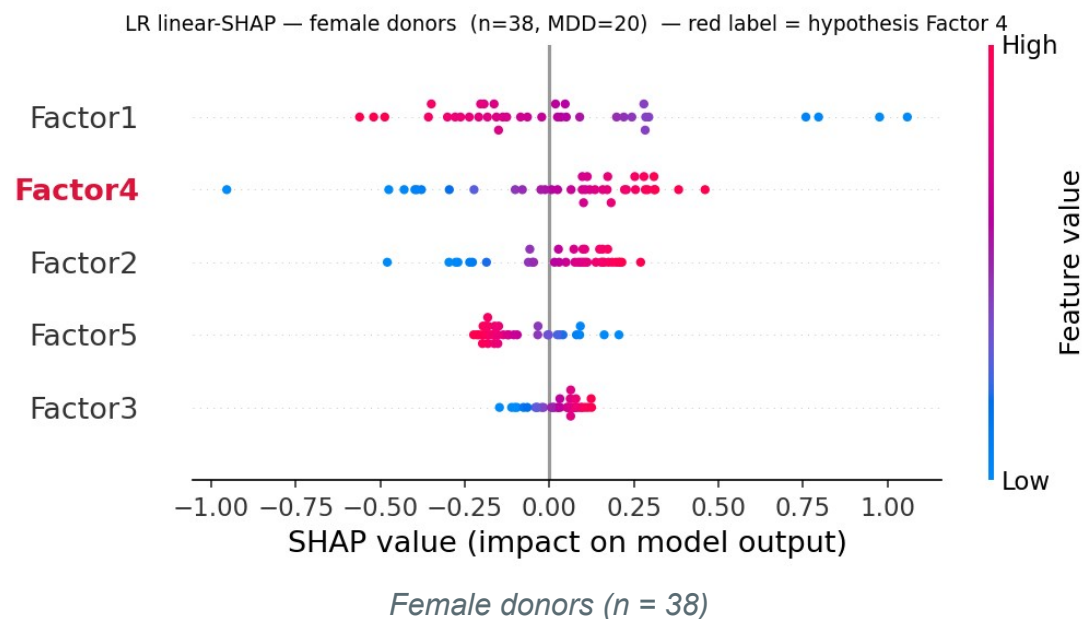
AUC · females · p=0.42



Communication-only LR SHAP (all donors). Factor 4 in red.

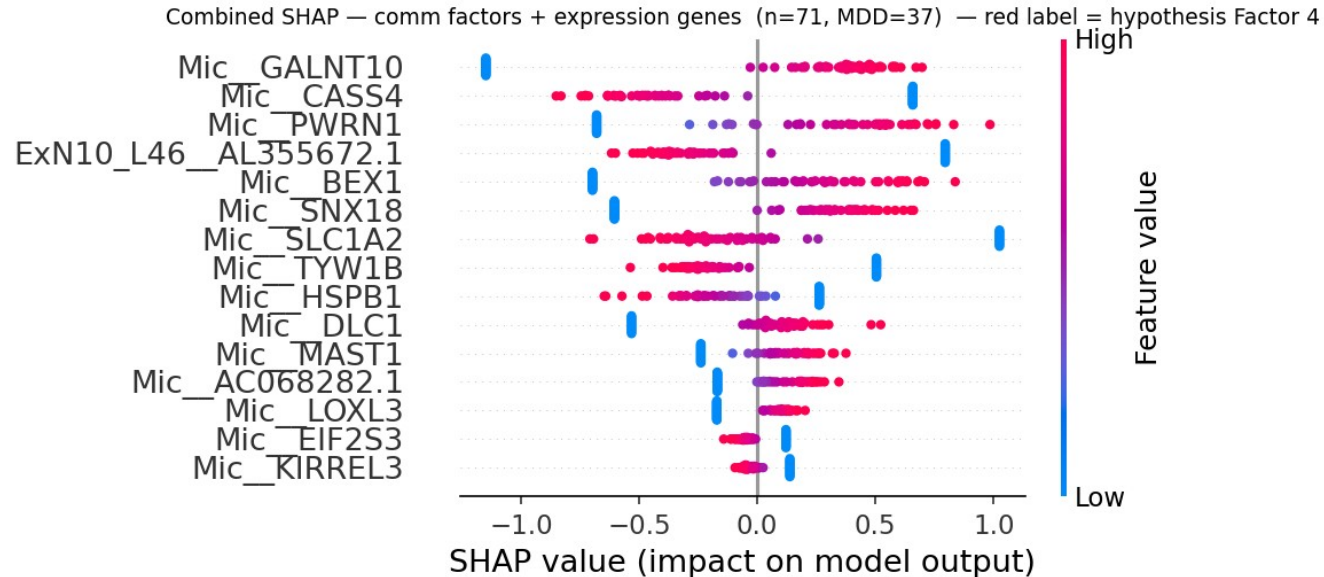
Standalone, the program discriminates MDD with AUC 0.632, concentrated in **males**. SHAP ranks Factor 4 **#2 of 5** — comparably across sexes (F 0.234, M 0.227); high values push the prediction toward MDD.

Factor 4 carries weight in females and males alike



The hypothesis program is not a male-only artefact: Factor 4 (red) sits among the top contributors in both sexes, with high values on the MDD side — the signature of a **shared axis**, not two unrelated findings.

When factors compete with genes, genes win

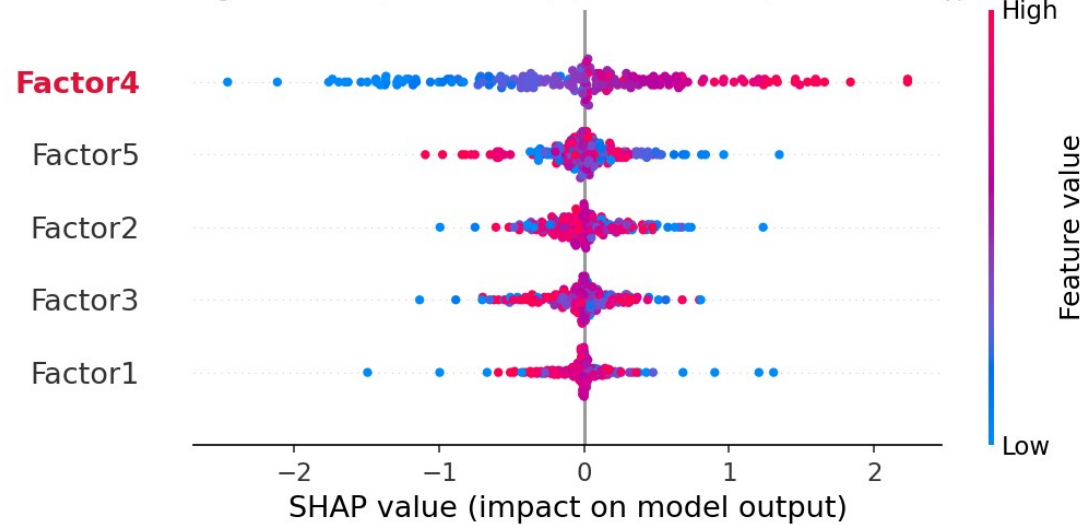


Combined-space SHAP: mRMR keeps 15 features — almost all microglial genes (Mic_). Every communication factor is dropped.*

Given variance-top-1000 genes + 5 factors, feature selection keeps **15 genes and discards every factor**. This is the feature-level reason combined adds no value: the program is a **low-rank summary of microglial state**, captured at least as well by the raw Mic genes.

Remove the leak, and Factor 4 leads

In-fold SHAP — leakage-safe factors (held-out donors) (n=213, MDD=111) — red label = hypothesis Factor 4



In-fold (leakage-safe) factors on held-out donors, pooled across 3 CV rounds (71 × 3 = 213).

0.649

held-out AUC — matches the rigorous 0.633, validating the in-fold path

Factor 4 → #1 communication factor

mean|SHAP| 0.610, again stronger in males (0.651) than females (0.575). Removing the mild all-donor leak makes the axis **more** prominent, not less.

So the biology is robust to the strictest leakage control.

DISCUSSION

A coherent, two-part answer

✓ THE BIOLOGY IS SUPPORTED

Factor 4 is the dominant, reproducible communication program.

It self-separates from the batch confound, leans toward MDD in both sexes, and becomes the #1 communication feature under leakage-safe estimation.

Directionally supports the shared microglia → neuron axis. L–R drivers (synaptic, glutamatergic, Ca^{2+} ; CACNA1C, IL1RAPL1) are biologically sensible.

— BUT IT ADDS NO PREDICTIVE VALUE

When factors and genes compete, mRMR discards every factor; combined never CI-separates from expression.

This is expected, not paradoxical: communication is derived from expression. A rank-5 program is a lossy summary of the same microglial state the raw genes encode at full resolution.

At $n = 71$ the compression buys interpretability, not discriminative power.

The signal is real, but not independent.

CONCLUSION

A qualified yes — honest by design

Modelling cell–cell communication recovers a single, reproducible microglia → ExN10_L46 program, elevated in MDD donors of both sexes and the most important communication feature under leakage-safe estimation — **directionally supporting** the shared-axis hypothesis. But it carries **no predictive information independent of microglial expression** at $n = 71$: when the two compete, expression wins.

The shared axis is visible and real — a low-rank shadow of microglial transcriptional state, not an independent, value-adding signal.

KEY LIMITATIONS

n = 71 — wide CIs; a +0.05 AUC value-add is not resolvable (and why no GNN).

Sex = cohort

— no clean cross-sex test; within-sex value-add instead.

Factor 4 ≠ pure microglia

— top sender is Endothelial (0.79); vascular + immune → neuron.

Trend-level

— $p = 0.057$ overall, $p = 0.05$ in males (borderline).

Thank you

Questions?

Giorgos Boulogeorgos · Andreas Mici | MLCB Team Project · NKUA, MSc Data Science and Information Technologies · Supervisor: Prof. E. Manolakos · June 2026